



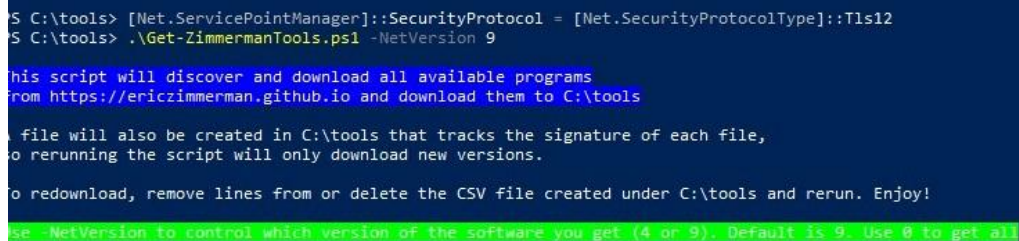
Cyberster Blue Team
Internship
WEEK 08 Report

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CSI-B1-487

Windows Forensic Artifact Analysis

Introduction

The focus of Weeks 8 and 9 was to perform a deep-dive forensic audit of a Windows workstation to reconstruct user activity and identify potential anti-forensic measures. The primary objective was to transition from event log auditing to analyzing persistent artifacts—specifically **Prefetch**, **Recycle Bin**, and **Thumbcache**—using the Zimmerman toolset to extract actionable intelligence.



```
S C:\tools> [Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12
S C:\tools> .\Get-ZimmermanTools.ps1 -NetVersion 9

This script will discover and download all available programs
from https://ericzimmerman.github.io and download them to C:\tools

A file will also be created in C:\tools that tracks the signature of each file,
so rerunning the script will only download new versions.

To redownload, remove lines from or delete the CSV file created under C:\tools and rerun. Enjoy!

Use -NetVersion to control which version of the software you get (4 or 9). Default is 9. Use 0 to get all
```

Day 53-55: Advanced Event Log Auditing & Authentication Baseline

Establish a human-activity baseline and identify anomalous authentication attempts through security event logs.

- **Authentication Mapping:** I utilized **EvtxECmd** to parse security logs, specifically tracking **Event ID 4624**. I successfully differentiated between **Logon Type 2** (Interactive/Physical) and **Logon Type 5** (Service/System), allowing for the isolation of actual human working hours.

Event log details
Flags: IsDirty
Chunk count: 320
Stored/Calculated CRC: CDEEBF7E/CDEEBF7E
Earliest timestamp: 2026-05-07 18:31:08.1700073
Latest timestamp: 2026-05-10 05:54:59.0673650
Total event log records found: 28,612

Records included: 28,612 Errors: 0 Events dropped: 0

Metrics (including dropped events)

Event ID	Count
1101	6
4608	6
4616	11
4624	3,531
4625	9
4634	10
4648	119
4672	3,502
4688	65
4696	6
4797	10
4798	7,056
4799	240
4826	6
4902	6
5024	6
5033	6
5058	1,396
5059	705
5061	1,372
5379	10,510
5382	28
6406	6

Processed 1 file in 21.6503 seconds

PS C:\tools\EvtxECmd>

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	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	RecordNum	EventRecon	TimeCreated	EventId	Level	Provider	Channel	ProcessId	ThreadId	Computer	ChunkNum	UserId	MapDescrip	UserName	RemoteHost	PayloadData
25	3897121	3897121	2026-05-08 14:5	4624	LogAlways	Microsoft-Windo Security		776	1132	DESKTOP-2HO		0	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
41	3897137	3897137	2026-05-08 14:5	4624	LogAlways	Microsoft-Windo Security		776	1132	DESKTOP-2HO		0	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
166	3897261	3897261	2026-05-08 14:5	4624	LogAlways	Microsoft-Windo Security		776	1132	DESKTOP-2HO		1	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
314	3897410	3897410	2026-05-08 14:5	4624	LogAlways	Microsoft-Windo Security		776	1088	DESKTOP-2HO		2	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
373	3897469	3897469	2026-05-08 14:5	4624	LogAlways	Microsoft-Windo Security		776	1124	DESKTOP-2HO		3	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
375	3897471	3897471	2026-05-08 14:5	4624	LogAlways	Microsoft-Windo Security		776	1124	DESKTOP-2HO		3	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
378	3897474	3897474	2026-05-08 14:5	4624	LogAlways	Microsoft-Windo Security		776	2398	DESKTOP-2HO		3	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
387	3897483	3897483	2026-05-08 14:5	4624	LogAlways	Microsoft-Windo Security		776	1124	DESKTOP-2HO		3	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
447	3897543	3897543	2026-05-08 14:5	4624	LogAlways	Microsoft-Windo Security		776	1132	DESKTOP-2HO		4	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
449	3897545	3897545	2026-05-08 14:5	4624	LogAlways	Microsoft-Windo Security		776	16776	DESKTOP-2HO		4	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
452	3897548	3897548	2026-05-08 14:5	4624	LogAlways	Microsoft-Windo Security		776	1088	DESKTOP-2HO		4	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
454	3897550	3897550	2026-05-08 14:5	4624	LogAlways	Microsoft-Windo Security		776	1132	DESKTOP-2HO		4	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
474	3897570	3897570	2026-05-08 14:5	4624	LogAlways	Microsoft-Windo Security		776	16776	DESKTOP-2HO		4	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
476	3897572	3897572	2026-05-08 14:5	4624	LogAlways	Microsoft-Windo Security		776	16776	DESKTOP-2HO		4	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
483	3897576	3897576	2026-05-08 14:5	4624	LogAlways	Microsoft-Windo Security		776	2332	DESKTOP-2HO		4	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
485	3897579	3897579	2026-05-08 14:5	4624	LogAlways	Microsoft-Windo Security		776	1124	DESKTOP-2HO		4	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
487	3897583	3897583	2026-05-08 15:0	4624	LogAlways	Microsoft-Windo Security		776	16776	DESKTOP-2HO		4	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
488	3897581	3897581	2026-05-08 15:0	4624	LogAlways	Microsoft-Windo Security		776	1132	DESKTOP-2HO		4	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
502	3897586	3897586	2026-05-08 15:0	4624	LogAlways	Microsoft-Windo Security		776	2332	DESKTOP-2HO		5	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
504	3897600	3897600	2026-05-08 15:0	4624	LogAlways	Microsoft-Windo Security		776	1124	DESKTOP-2HO		5	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
513	3897609	3897609	2026-05-08 15:0	4624	LogAlways	Microsoft-Windo Security		776	1124	DESKTOP-2HO		5	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
515	3897611	3897611	2026-05-08 15:0	4624	LogAlways	Microsoft-Windo Security		776	1132	DESKTOP-2HO		5	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
570	3897672	3897672	2026-05-08 15:0	4624	LogAlways	Microsoft-Windo Security		776	1132	DESKTOP-2HO		5	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
585	3897681	3897681	2026-05-08 15:0	4624	LogAlways	Microsoft-Windo Security		776	1124	DESKTOP-2HO		5	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log
588	3897684	3897684	2026-05-08 15:0	4624	LogAlways	Microsoft-Windo Security		776	1124	DESKTOP-2HO		5	Successful logon	WORKGROUP\PI - (-)		Target: NT AUTH Log

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	D	E	F	G	H	I	J	K	L	M	N	O	P	Q		
1	EventId	Level	Provider	Channel	ProcessId	ThreadId	Computer	ChunkNum	UserId	MapDescription	UserName	RemoteHost	PayloadData1	PayloadData2	Payload	
4793_35	4624	LogAlways	Microsoft-Windows:Security		608	1135	DESKTOP-2HO	51		Successful logon	WORKGROUP\PI - (-)		Target: Font Driver Host\UMFD-0	LogonType 2	LogonId	
4795_35	4624	LogAlways	Microsoft-Windows:Security		608	1096	DESKTOP-2HO	51		Successful logon	WORKGROUP\PI - (-)		Target: Font Driver Host\UMFD-0	LogonType 2	LogonId	
4797_35	4624	LogAlways	Microsoft-Windows:Security		608	1060	DESKTOP-2HO	51		Successful logon	WORKGROUP\PI - (-)		Target: Window Manager\DWIM-1	LogonType 2	LogonId	
4798_35	4624	LogAlways	Microsoft-Windows:Security		608	1060	DESKTOP-2HO	51		Successful logon	WORKGROUP\PI - (-)		Target: Window Manager\DWIM-1	LogonType 2	LogonId	
4814_00	4624	LogAlways	Microsoft-Windows:Security		608	1132	DESKTOP-2HO	53		Successful logon	WORKGROUP\PI DESKTOP-2HO	Target: DESKTOP-2HO\GEG\PointStar	LogonType 2	LogonId		
4815_00	4624	LogAlways	Microsoft-Windows:Security		608	1132	DESKTOP-2HO	53		Successful logon	WORKGROUP\PI DESKTOP-2HO	Target: DESKTOP-2HO\GEG\PointStar	LogonType 2	LogonId		
4933_37	4624	LogAlways	Microsoft-Windows:Security		628	1148	DESKTOP-2HO	71		Successful logon	WORKGROUP\PI - (-)		Target: Font Driver Host\UMFD-0	LogonType 2	LogonId	
4935_37	4624	LogAlways	Microsoft-Windows:Security		628	1068	DESKTOP-2HO	71		Successful logon	WORKGROUP\PI - (-)		Target: Font Driver Host\UMFD-0	LogonType 2	LogonId	
4936_38	4624	LogAlways	Microsoft-Windows:Security		628	1104	DESKTOP-2HO	72		Successful logon	WORKGROUP\PI - (-)		Target: Window Manager\DWIM-1	LogonType 2	LogonId	
4937_38	4624	LogAlways	Microsoft-Windows:Security		628	1104	DESKTOP-2HO	72		Successful logon	WORKGROUP\PI - (-)		Target: Window Manager\DWIM-1	LogonType 2	LogonId	
4952_44	4624	LogAlways	Microsoft-Windows:Security		628	1144	DESKTOP-2HO	72		Successful logon	WORKGROUP\PI DESKTOP-2HO	Target: DESKTOP-2HO\GEG\PointStar	LogonType 2	LogonId		
4953_44	4624	LogAlways	Microsoft-Windows:Security		628	1144	DESKTOP-2HO	72		Successful logon	WORKGROUP\PI DESKTOP-2HO	Target: DESKTOP-2HO\GEG\PointStar	LogonType 2	LogonId		
4977_38	4624	LogAlways	Microsoft-Windows:Security		628	3228	DESKTOP-2HO	85		Successful logon	WORKGROUP\PI DESKTOP-2HO	Target: DESKTOP-2HO\GEG\PointStar	LogonType 2	LogonId		
4978_38	4624	LogAlways	Microsoft-Windows:Security		628	3228	DESKTOP-2HO	85		Successful logon	WORKGROUP\PI DESKTOP-2HO	Target: DESKTOP-2HO\GEG\PointStar	LogonType 2	LogonId		
5069_07	4624	LogAlways	Microsoft-Windows:Security		628	18903	DESKTOP-2HO	110		Successful logon	WORKGROUP\PI DESKTOP-2HO	Target: DESKTOP-2HO\GEG\PointStar	LogonType 2	LogonId		
5069_07	4624	LogAlways	Microsoft-Windows:Security		628	18903	DESKTOP-2HO	110		Successful logon	WORKGROUP\PI DESKTOP-2HO	Target: DESKTOP-2HO\GEG\PointStar	LogonType 2	LogonId		
10397_02	4624	LogAlways	Microsoft-Windows:Security		592	1044	DESKTOP-2HO	115		Successful logon	WORKGROUP\PI - (-)		Target: Font Driver Host\UMFD-0	LogonType 2	LogonId	
10393_03	4624	LogAlways	Microsoft-Windows:Security		592	758	DESKTOP-2HO	115		Successful logon	WORKGROUP\PI - (-)		Target: Font Driver Host\UMFD-0	LogonType 2	LogonId	
10370_03	4624	LogAlways	Microsoft-Windows:Security		592	758	DESKTOP-2HO	115		Successful logon	WORKGROUP\PI - (-)		Target: Window Manager\DWIM-1	LogonType 2	LogonId	
10380_03	4624	LogAlways	Microsoft-Windows:Security		592	758	DESKTOP-2HO	115		Successful logon	WORKGROUP\PI - (-)		Target: Window Manager\DWIM-1	LogonType 2	LogonId	
10458_11	4624	LogAlways	Microsoft-Windows:Security		592	758	DESKTOP-2HO	116		Successful logon	WORKGROUP\PI DESKTOP-2HO	Target: DESKTOP-2HO\GEG\PointStar	LogonType 2	LogonId		
10460_11	4624	LogAlways	Microsoft-Windows:Security		592	758	DESKTOP-2HO	116		Successful logon	WORKGROUP\PI DESKTOP-2HO	Target: DESKTOP-2HO\GEG\PointStar	LogonType 2	LogonId		
12703_49	4624	LogAlways	Microsoft-Windows:Security		576	1128	DESKTOP-2HO	152		Successful logon	WORKGROUP\PI - (-)		Target: Font Driver Host\UMFD-0	LogonType 2	LogonId	
12705_49	4624	LogAlways	Microsoft-Windows:Security		576	1120	DESKTOP-2HO	152		Successful logon	WORKGROUP\PI - (-)		Target: Font Driver Host\UMFD-0	LogonType 2	LogonId	
13302_49	4624	LogAlways	Microsoft-Windows:Security		576	1120	DESKTOP-2HO	152		Successful logon	WORKGROUP\PI - (-)		Target: Window Manager\DWIM-1	LogonType 2	LogonId	

- **Threat Hunting (Failed Logons):** A retrospective search for **Event ID 4625** (Failed Logon) was performed. I documented a specific failure on **May 10th**, verifying the system's ability to log unauthorized access attempts.

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	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	RecordNum	EventRecon	TimeCreated	EventId	Level	Provider	Channel	ProcessId	ThreadId	Computer	ChunkNum	UserId	MapDescrip	UserName	RemoteHost	PayloadData
1265	3910151	3910151	2026-05-09 11:2	4625	LogAlways	Microsoft-Windo Security		592	9328	DESKTOP-2HO		143	Failed logon	-	(192.168.18.23 Target: v	Log
1266	3910152	3910152	2026-05-09 11:2	4625	LogAlways	Microsoft-Windo Security		592	9328	DESKTOP-2HO		143	Failed logon	-	(192.168.18.23 Target: v	Log
1267	3910153	3910153	2026-05-09 11:2	4625	LogAlways	Microsoft-Windo Security		592	9328	DESKTOP-2HO		143	Failed logon	-	(192.168.18.23 Target: v	Log
14054	3911150	3911150	2026-05-10 05:4	4625	LogAlways	Microsoft-Windo Security		576	4964	DESKTOP-2HO		155	Failed logon	-	(192.168.18.23 Target: v	Log
14055	3911151	3911151	2026-05-10 05:4	4625	LogAlways	Microsoft-Windo Security		576	4964	DESKTOP-2HO		155	Failed logon	-	(192.168.18.23 Target: v	Log
14056	3911152	3911152	2026-05-10 05:4	4625	LogAlways	Microsoft-Windo Security		576	4964	DESKTOP-2HO		155	Failed logon	-	(192.168.18.23 Target: v	Log
2738	3895842	3895842	2026-05-08 14:5	4625	LogAlways	Microsoft-Windo Security		776	1132	DESKTOP-2HO		307	Failed logon	-	(192.168.18.23 Target: v	Log
2739	3895843	3895843	2026-05-08 14:5	4625	LogAlways	Microsoft-Windo Security		776	1132	DESKTOP-2HO		307	Failed logon	-	(192.168.18.23 Target: v	Log
2740	3895844	3895844	2026-05-08 14:5	4625	LogAlways	Microsoft-Windo Security		776	1132	DESKTOP-2HO		307	Failed logon	-	(192.168.18.23 Target: v	Log

- **Privilege Audit:** I monitored **Event ID 4672** to verify when administrative privileges were assigned. This confirmed that forensic tools were executed with the necessary elevated permissions for artifact acquisition.

My_Security_Analysis

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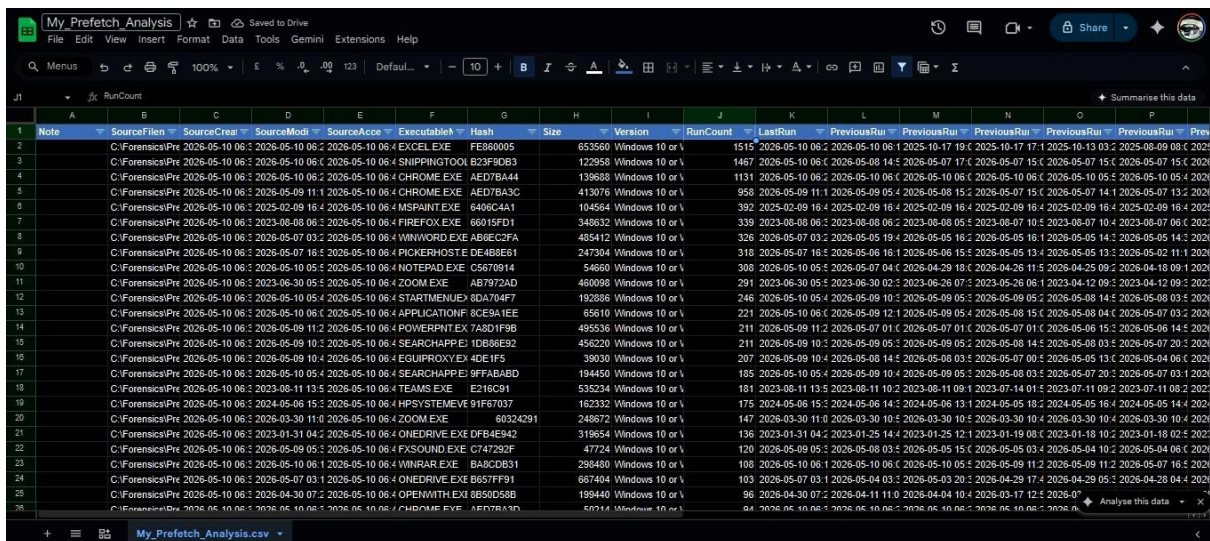
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RecordNumber	EventRecordId	TimeCreated	EventId	Level	Provider	Channel	ProcessId	ThreadId	Computer	ChunkNum	UserId	MapDescription	UserName	RemoteHost	Payload
3897122	3897122	2026-05-08 14:53:39	4672	LogAlways	Microsoft-Windows-Security		776	1132	DESKTOP-2HO	0		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897138	3897138	2026-05-08 14:54:10	4672	LogAlways	Microsoft-Windows-Security		776	1132	DESKTOP-2HO	0		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897262	3897262	2026-05-08 14:54:41	4672	LogAlways	Microsoft-Windows-Security		776	1132	DESKTOP-2HO	1		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897411	3897411	2026-05-08 14:55:15	4672	LogAlways	Microsoft-Windows-Security		776	1088	DESKTOP-2HO	3		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897470	3897470	2026-05-08 14:55:58	4672	LogAlways	Microsoft-Windows-Security		776	1124	DESKTOP-2HO	3		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897472	3897472	2026-05-08 14:55:59	4672	LogAlways	Microsoft-Windows-Security		776	1124	DESKTOP-2HO	3		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897475	3897475	2026-05-08 14:56:11	4672	LogAlways	Microsoft-Windows-Security		776	22596	DESKTOP-2HO	3		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897484	3897484	2026-05-08 14:56:31	4672	LogAlways	Microsoft-Windows-Security		776	1124	DESKTOP-2HO	3		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897544	3897544	2026-05-08 14:57:02	4672	LogAlways	Microsoft-Windows-Security		776	1124	DESKTOP-2HO	4		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897546	3897546	2026-05-08 14:57:33	4672	LogAlways	Microsoft-Windows-Security		776	16776	DESKTOP-2HO	4		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897549	3897549	2026-05-08 14:57:47	4672	LogAlways	Microsoft-Windows-Security		776	1088	DESKTOP-2HO	4		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897551	3897551	2026-05-08 14:58:03	4672	LogAlways	Microsoft-Windows-Security		776	1132	DESKTOP-2HO	4		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897571	3897571	2026-05-08 14:58:25	4672	LogAlways	Microsoft-Windows-Security		776	16776	DESKTOP-2HO	4		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897573	3897573	2026-05-08 14:58:35	4672	LogAlways	Microsoft-Windows-Security		776	16776	DESKTOP-2HO	4		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897577	3897577	2026-05-08 14:59:05	4672	LogAlways	Microsoft-Windows-Security		776	2332	DESKTOP-2HO	4		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897580	3897580	2026-05-08 14:59:36	4672	LogAlways	Microsoft-Windows-Security		776	1124	DESKTOP-2HO	4		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897584	3897584	2026-05-08 15:00:07	4672	LogAlways	Microsoft-Windows-Security		776	16776	DESKTOP-2HO	4		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897592	3897592	2026-05-08 15:00:37	4672	LogAlways	Microsoft-Windows-Security		776	1132	DESKTOP-2HO	4		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897599	3897599	2026-05-08 15:01:08	4672	LogAlways	Microsoft-Windows-Security		776	2332	DESKTOP-2HO	5		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897601	3897601	2026-05-08 15:01:19	4672	LogAlways	Microsoft-Windows-Security		776	1124	DESKTOP-2HO	5		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897610	3897610	2026-05-08 15:01:38	4672	LogAlways	Microsoft-Windows-Security		776	1124	DESKTOP-2HO	5		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897612	3897612	2026-05-08 15:02:09	4672	LogAlways	Microsoft-Windows-Security		776	1132	DESKTOP-2HO	5		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897673	3897673	2026-05-08 15:02:39	4672	LogAlways	Microsoft-Windows-Security		776	1132	DESKTOP-2HO	5		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897682	3897682	2026-05-08 15:03:10	4672	LogAlways	Microsoft-Windows-Security		776	1124	DESKTOP-2HO	5		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	
3897685	3897685	2026-05-08 15:03:25	4672	LogAlways	Microsoft-Windows-Security		776	1124	DESKTOP-2HO	5		Administrative logon	NT AUTHORITY\SYSTEM	(S-1-5-Privilege	

Day 56-57: Program Execution Analysis (Prefetch)

To identify the most frequently executed applications and establish a timeline of tool usage.

- **Artifact Extraction:** Utilizing **PECmd**, I parsed the **C:\Windows\Prefetch** directory. This provided a "record of intent," proving exactly which applications were executed, regardless of whether they were later deleted.
- **Top 20 Execution Audit:** I generated a prioritized table of program executions. The analysis revealed **EXCEL.EXE** (1,515 runs) and **SNIPPINGTOOL.EXE** (1,467 runs) as the primary tools, consistent with documented internship reporting tasks.



Note	SourceFile	SourceCrea	SourceModi	SourceAcce	Executable	Hash	Size	Version	RunCount	LastRun	PreviousRun	PreviousRun	PreviousRun	PreviousRun	PreviousRun	PreviousRun
1	C:\Forensics\Pre	2026-05-10 06:2	2026-05-10 06:2	2026-05-10 06:2	EXCEL.EXE	F8880005	653580	Windows 10 or v	1515	2026-05-10 06:2	2026-05-10 06:1	2026-05-10 17:1	2026-05-10 17:1	2026-05-10 17:1	2026-05-10 17:1	2026-05-10 17:1
2	C:\Forensics\Pre	2026-05-10 06:2	2026-05-10 06:2	2026-05-10 06:2	SNIPPINGTOOL	E29F5D03	122950	Windows 10 or v	1467	2026-05-10 06:2	2026-05-10 14:2	2026-05-07 17:2	2026-05-07 15:2	2026-05-07 15:2	2026-05-07 15:2	2026-05-07 15:2
3	C:\Forensics\Pre	2026-05-10 06:2	2026-05-10 06:2	2026-05-10 06:2	CHROME.EXE	AED7BA44	138680	Windows 10 or v	1131	2026-05-10 06:2	2026-05-10 06:2	2026-05-10 06:2	2026-05-10 06:2	2026-05-10 06:2	2026-05-10 06:2	2026-05-10 06:2
4	C:\Forensics\Pre	2026-05-10 06:2	2026-05-09 11:1	2026-05-10 06:2	CHROME.EXE	AED7BA3C	413070	Windows 10 or v	950	2026-05-09 11:1	2026-05-09 05:4	2026-05-08 15:2	2026-05-07 15:2	2026-05-07 14:1	2026-05-07 13:2	2026-05-07 13:2
5	C:\Forensics\Pre	2026-05-10 06:2	2026-02-09 16:4	2026-05-10 06:2	MSPAINT.EXE	6409C441	104564	Windows 10 or v	392	2026-02-09 16:4	2026-02-09 16:4	2026-02-09 16:4	2026-02-09 16:4	2026-02-09 16:4	2026-02-09 16:4	2026-02-09 16:4
6	C:\Forensics\Pre	2026-05-10 06:2	2026-08-08 06:2	2026-05-10 06:2	FIREFOX.EXE	66015FD1	348632	Windows 10 or v	339	2026-08-08 06:2	2026-08-08 06:2	2026-08-08 06:2	2026-08-07 10:4	2026-08-07 10:4	2026-08-07 06:2	2026-08-07 06:2
7	C:\Forensics\Pre	2026-05-10 06:2	2026-05-07 03:2	2026-05-10 06:2	WINWORD.EXE	AB8EC2FA	488412	Windows 10 or v	326	2026-05-07 03:2	2026-05-06 19:4	2026-05-05 16:2	2026-05-05 16:1	2026-05-05 14:2	2026-05-05 14:2	2026-05-05 14:2
8	C:\Forensics\Pre	2026-05-10 06:2	2026-05-07 16:2	2026-05-10 06:2	PICKERHOSTE	DE4B8E61	247304	Windows 10 or v	318	2026-05-07 16:2	2026-05-06 16:1	2026-05-06 15:2	2026-05-05 13:4	2026-05-05 13:3	2026-05-05 11:1	2026-05-05 11:1
9	C:\Forensics\Pre	2026-05-10 06:2	2026-05-10 05:2	2026-05-10 06:2	NOTEPAD.EXE	C5670914	44660	Windows 10 or v	308	2026-05-10 05:2	2026-05-07 04:2	2026-04-29 18:2	2026-04-26 11:5	2026-04-25 09:2	2026-04-18 09:1	2026-04-18 09:1
10	C:\Forensics\Pre	2026-05-10 06:2	2026-06-30 05:2	2026-05-10 06:2	ZOOM.EXE	AB7972AD	460098	Windows 10 or v	291	2026-06-30 05:2	2026-06-30 02:2	2026-06-26 07:2	2026-05-26 06:1	2026-04-12 09:2	2026-04-12 09:2	2026-04-12 09:2
11	C:\Forensics\Pre	2026-05-10 06:2	2026-05-10 05:4	2026-05-10 06:2	STARTMENUE	8DA704F7	192886	Windows 10 or v	246	2026-05-10 05:4	2026-05-09 10:2	2026-05-09 05:2	2026-05-09 05:2	2026-05-08 14:2	2026-05-08 03:2	2026-05-08 03:2
12	C:\Forensics\Pre	2026-05-10 06:2	2026-05-10 06:2	2026-05-10 06:2	APPLICATIONF	8CE9A1EE	65610	Windows 10 or v	221	2026-05-10 06:2	2026-05-09 12:1	2026-05-09 05:4	2026-05-08 15:2	2026-05-08 04:2	2026-05-07 03:2	2026-05-07 03:2
13	C:\Forensics\Pre	2026-05-10 06:2	2026-05-09 11:2	2026-05-10 06:2	POWERPNTX	7A8D1F98	495536	Windows 10 or v	211	2026-05-09 11:2	2026-05-07 01:2	2026-05-07 01:2	2026-05-07 01:2	2026-05-06 15:2	2026-05-06 14:2	2026-05-06 14:2
14	C:\Forensics\Pre	2026-05-10 06:2	2026-05-09 10:2	2026-05-10 06:2	SEARCHAPPE	1DB8E892	456220	Windows 10 or v	211	2026-05-09 10:2	2026-05-09 05:2	2026-05-09 05:2	2026-05-08 14:2	2026-05-08 03:2	2026-05-07 20:2	2026-05-07 20:2
15	C:\Forensics\Pre	2026-05-10 06:2	2026-05-09 10:4	2026-05-10 06:2	EQUIPROXYE	4DE1F5	39030	Windows 10 or v	207	2026-05-09 10:4	2026-05-08 14:2	2026-05-08 03:2	2026-05-07 00:2	2026-05-05 13:2	2026-05-04 06:2	2026-05-04 06:2
16	C:\Forensics\Pre	2026-05-10 06:2	2026-05-10 05:4	2026-05-10 06:2	SEARCHAPPE	9FFABABD	194450	Windows 10 or v	185	2026-05-10 05:4	2026-05-09 10:4	2026-05-09 05:2	2026-05-08 03:2	2026-05-07 20:2	2026-05-07 03:2	2026-05-07 03:2
17	C:\Forensics\Pre	2026-05-10 06:2	2023-08-11 13:2	2026-05-10 06:2	TEAMS.EXE	E216C91	535234	Windows 10 or v	181	2023-08-11 13:2	2023-08-11 10:2	2023-08-11 09:1	2023-07-14 01:2	2023-07-11 09:2	2023-07-11 08:2	2023-07-11 08:2
18	C:\Forensics\Pre	2026-05-10 06:2	2024-05-06 15:2	2026-05-10 06:2	HPV\STEVEVE	91F67037	162332	Windows 10 or v	175	2024-05-06 15:2	2024-05-06 14:2	2024-05-06 13:1	2024-05-05 18:2	2024-05-05 16:2	2024-05-05 14:2	2024-05-05 14:2
19	C:\Forensics\Pre	2026-05-10 06:2	2026-03-30 11:0	2026-05-10 06:2	ZOOM.EXE	60324291	246672	Windows 10 or v	147	2026-03-30 11:0	2026-03-30 10:2	2026-03-30 10:2	2026-03-30 10:4	2026-03-30 10:4	2026-03-30 10:4	2026-03-30 10:4
20	C:\Forensics\Pre	2026-05-10 06:2	2023-01-31 04:2	2026-05-10 06:2	ONEDRIVE	DFB4E94F	318654	Windows 10 or v	136	2023-01-31 04:2	2023-01-25 14:2	2023-01-25 12:1	2023-01-18 06:2	2023-01-18 10:2	2023-01-18 02:2	2023-01-18 02:2
21	C:\Forensics\Pre	2026-05-10 06:2	2026-05-09 05:2	2026-05-10 06:2	FXSOUND	EXE C747292F	47724	Windows 10 or v	120	2026-05-09 05:2	2026-05-08 03:2	2026-05-05 15:2	2026-05-05 03:4	2026-05-04 10:2	2026-05-04 06:2	2026-05-04 06:2
22	C:\Forensics\Pre	2026-05-10 06:2	2026-05-10 06:1	2026-05-10 06:2	WINRAR	EXE B48CDB31	298480	Windows 10 or v	108	2026-05-10 06:1	2026-05-10 06:2	2026-05-10 05:2	2026-05-09 11:2	2026-05-09 11:2	2026-05-07 16:2	2026-05-07 16:2
23	C:\Forensics\Pre	2026-05-10 06:2	2026-05-07 03:1	2026-05-10 06:2	ONEDRIVE	EXE B657F9F1	667404	Windows 10 or v	103	2026-05-07 03:1	2026-05-04 03:1	2026-05-03 20:1	2026-04-29 17:2	2026-04-29 05:2	2026-04-28 04:2	2026-04-28 04:2
24	C:\Forensics\Pre	2026-05-10 06:2	2026-04-30 07:2	2026-05-10 06:2	OPENDRIVE	EXE 8B59D58B	199440	Windows 10 or v	96	2026-04-30 07:2	2026-04-11 11:0	2026-04-04 10:4	2026-03-17 12:2	2026-03-17 12:2	2026-03-17 12:2	2026-03-17 12:2
25	C:\Forensics\Pre	2026-05-10 06:2	2026-05-10 06:2	2026-05-10 06:2	CHROME	EXE AED7BA3D	507114	Windows 10 or v	94	2026-05-10 06:2	2026-05-10 06:2	2026-05-10 06:2	2026-05-10 06:2	2026-05-10 06:2	2026-05-10 06:2	2026-05-10 06:2

- **Timeline Reconstruction:** The analysis identified a high-activity cluster on **May 10th**, correlating with the active forensic investigation and report generation phases.

Day 58-59: Data Deletion & Persistence (Recycle Bin & Thumbcache)

To recover metadata from deleted files and identify visual "ghost" evidence of purged assets.

- **Recycle Bin Recovery (RBCmd):** Due to system-level protections on `C:\$Recycle.Bin`, I performed a live acquisition using **robocopy** to move artifacts to a forensic directory.
- **Metadata Extraction:** I successfully parsed the user's specific SID folder (-1002). The audit uncovered metadata for deleted **Office 365 configuration files** and a test directory named **"haha"**.
- **Proof of Validation:** The "haha" folder test confirmed the mapping of **\$I** (Index/Metadata) and **\$R** (Recovery/Data) files, proving the system correctly abstracts original filenames during deletion.

```

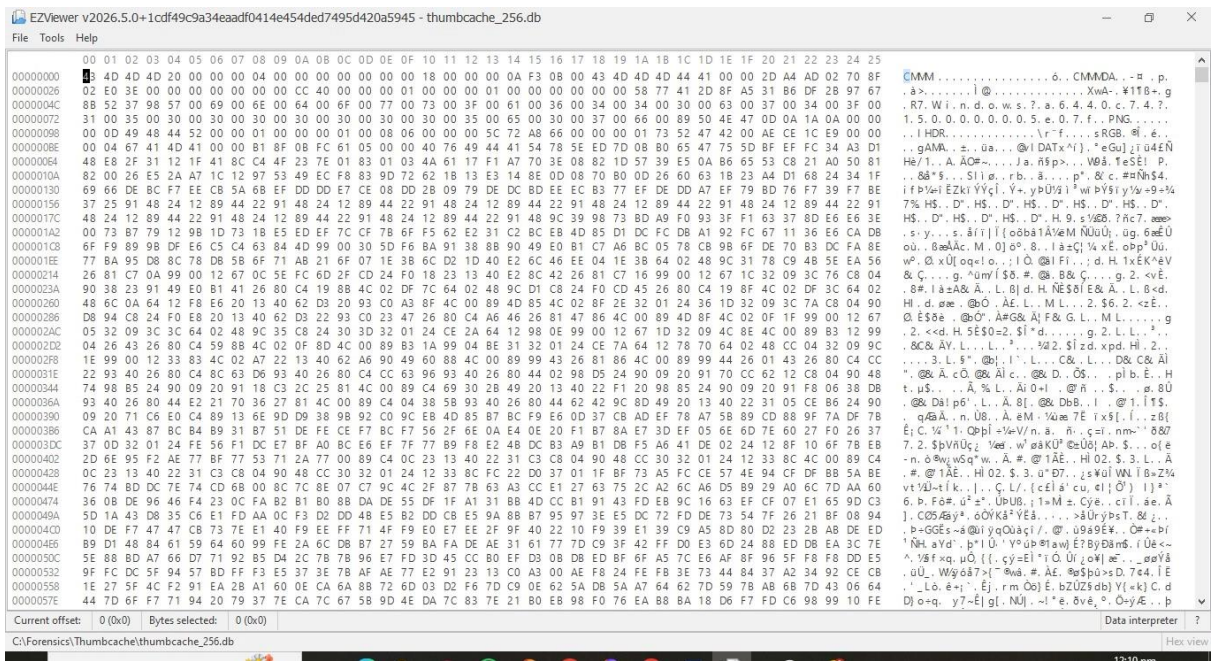
New Dir      0  C:\$Recycle.Bin\S-1-5-21-851654130-1114550027-1349266903-1002\SR1UFZL7\
New Dir      0  C:\$Recycle.Bin\S-1-5-21-851654130-1114550027-1349266903-1002\SR5YNQ95\
New Dir      4  C:\$Recycle.Bin\S-1-5-21-851654130-1114550027-1349266903-1002\SR5YNQ95\.idea\
100% New File      50  .gitignore
100% New File     277  misc.xml
100% New File     273  modules.xml
100% New File    1962  workspace.xml
New Dir      2  C:\$Recycle.Bin\S-1-5-21-851654130-1114550027-1349266903-1002\SR5YNQ95\src\
100% New File      74  Main.java
100% New File     440  Node.java
New Dir     15  C:\$Recycle.Bin\S-1-5-21-851654130-1114550027-1349266903-1002\SR5YNQ95\src\
100% New File    44207  ADD Anchor After.PNG
100% New File   28094  ADD Anchor Before.PNG
100% New File    18.4 m  BTS1.psd
100% New File    13.4 m  BTS2.psd
100% New File   19536  Convert Point After.PNG
100% New File   18257  Convert Point before.PNG
100% New File  319916  Curvature Full.PNG
100% New File   526785  Curvature Half.PNG
100% New File   36395  Delete Anchor After.PNG
100% New File   36797  Delete Anchor before.PNG
100% New File   31128  Freeform.PNG
100% New File   388038  Full Apple Standard.PNG
100% New File   442330  Half Apple Standard.PNG
100% New File     2.7 m  Pen Tool and It's Variants.pptx
100% New File     2.6 m  Samples.psd
New Dir      0  C:\$Recycle.Bin\S-1-5-21-851654130-1114550027-1349266903-1002\SRMTCNDL\
New Dir      1  C:\$Recycle.Bin\S-1-5-21-851654130-1114550027-1349266903-1002\SRMTCNDL\Riot Client.old\
100% New File    70.8 m  RiotClientServices.exe.old.old
New Dir      0  C:\$Recycle.Bin\S-1-5-21-851654130-1114550027-1349266903-1002\SRNR1I9U\
New Dir      3  C:\$Recycle.Bin\S-1-5-21-851654130-1114550027-1349266903-1002\SRPQZ6U5\
100% New File    1014  configuration-Office365-x64.xml
100% New File     296  Configuration.xml
100% New File     7.1 m  setup.exe
New Dir      1  C:\$Recycle.Bin\S-1-5-21-851654130-1114550027-1349266903-500\
100% New File    129  desktop.ini

-----
Total Copied Skipped Mismatch FAILED Extras
Dirs :   17   16     1      0      0      0
Files :  505  505     0      0      0      0
Bytes : 2.849 g 2.849 g     0      0      0      0
Times : 0:00:28 0:00:27     0      0  0:00:00 0:00:00

Speed :      110423436 Bytes/sec.
Speed :      6318.479 MegaBytes/min.
```

- **Thumbcache "Ghost" Hunt:** Using **EZViewer**, I inspected the `thumbcache_256.db`. I successfully identified visual thumbnails of images

that had been permanently deleted from the disk, demonstrating that visual traces persist in the user profile even after a Recycle Bin purge.



Report Summary and Final Verification

This report serves as a comprehensive technical audit of Windows forensic artifacts. By correlating **Event Logs** (Authentication), **Prefetch** (Execution), and **Recycle Bin/Thumbcache** (Deletion), I have established a verified timeline of user activity. The intelligence gathered confirms that while a user may attempt to delete data, persistent system artifacts provide a redundant and accurate record of activity, ensuring a stable and defensible forensic baseline.